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# Support Vector Machines for Alzheimer’s Disease Classification

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## Abstract

This report evaluates three Support Vector Machine (SVM) classifiers—linear, polynomial kernel, and Radial Basis Function (RBF) kernel—for distinguishing stable normal controls (sNC) from stable Dementia of Alzheimer’s Type patients (sDAT) using cortical glucose metabolism features derived from FDG-PET brain imaging. Hyperparameters were tuned via five-fold stratified cross-validation. The polynomial kernel SVM with  $C = 1.3895$  and degree  $d = 3$  achieved the highest overall accuracy (0.9069) and balanced accuracy (0.8935), outperforming both the linear and RBF models. These results indicate that moderate nonlinear relationships exist between cortical metabolic activity and DAT diagnosis.

## 1 Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder that affects cognitive function and memory. A commonly used research category for diagnosis is **Dementia of Alzheimer’s Type (DAT)**. Early identification of DAT using brain imaging biomarkers is important for improving diagnosis and understanding disease progression.

In this assignment, machine learning techniques were used to classify individuals as either **stable normal controls (sNC)** or **stable DAT patients (sDAT)** using glucose metabolism measurements obtained from FDG-PET brain imaging. Each subject is represented using metabolic activity from **14 cortical brain regions**.

Support Vector Machines (SVMs) were used for this classification task because they perform well in high-dimensional feature spaces and can model both linear and nonlinear relationships. Three SVM models were evaluated: a linear SVM, a polynomial kernel SVM, and a Radial Basis Function (RBF) kernel SVM. Performance was evaluated using accuracy, sensitivity, specificity, precision, recall, and balanced accuracy.

## 2 Linear SVM

A linear SVM classifier was trained using the provided training dataset. Prior to training, features were standardized using z-score normalization to ensure all features contributed equally to the model. The key hyperparameter for the linear SVM is the regularization parameter  $C$ , which controls the trade-off between maximizing the margin and minimizing classification errors. A grid search was performed over the range  $C \in [10^{-4}, 10^4]$ , with five-fold stratified cross-validation used to determine the optimal parameter based on balanced accuracy.

The optimal parameter obtained was  $C = 0.1083$ .

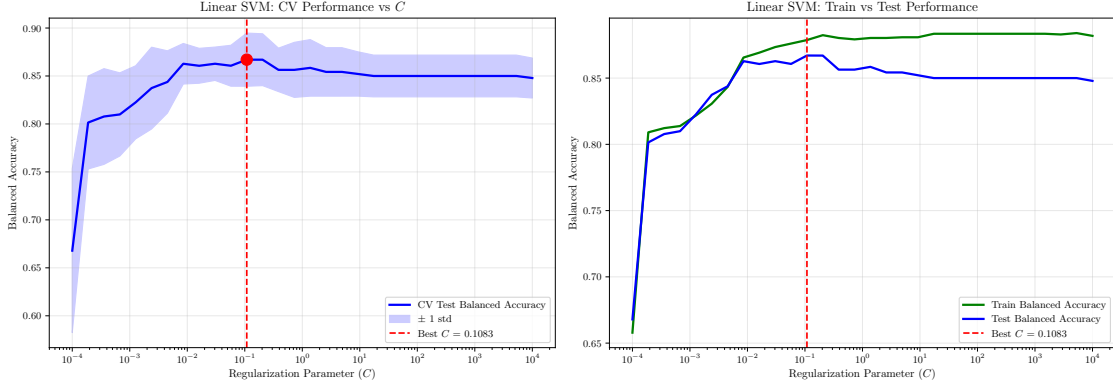


Figure 1: Cross-validation performance of the linear SVM for different values of the regularization parameter  $C$ .

### Test Performance

Table 1: Test set performance of the linear SVM.

| Metric            | Value  |
|-------------------|--------|
| Accuracy          | 0.8510 |
| Sensitivity       | 0.9344 |
| Specificity       | 0.8265 |
| Precision         | 0.6129 |
| Recall            | 0.9344 |
| Balanced Accuracy | 0.8805 |

The linear SVM achieved strong sensitivity, meaning that most DAT patients were correctly identified. However, the relatively lower precision indicates that some healthy controls were misclassified as DAT.

## 3 Polynomial Kernel SVM

To capture nonlinear relationships between cortical metabolic features and DAT classification, a polynomial kernel SVM was trained. The polynomial kernel is defined as

$$K(x_i, x_j) = (x_i^\top x_j + 1)^d.$$

Two hyperparameters were tuned: the regularization parameter  $C$  and the polynomial degree  $d$ . The search ranges were  $C \in [10^{-2}, 10^4]$  and  $d \in \{2, 3, 4, 5\}$ . Using cross-validation, the best parameters obtained were  $C = 1.3895$  and  $d = 3$ .

### Test Performance

The polynomial SVM achieved the highest overall accuracy and balanced accuracy among the tested models.

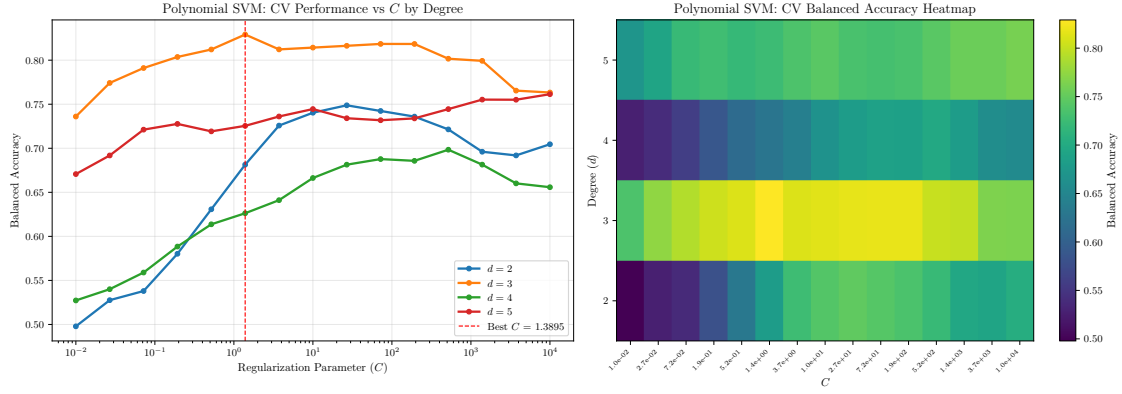


Figure 2: Cross-validation balanced accuracy for different polynomial kernel parameters.

Table 2: Test set performance of the polynomial kernel SVM.

| Metric            | Value  |
|-------------------|--------|
| Accuracy          | 0.9069 |
| Sensitivity       | 0.8689 |
| Specificity       | 0.9181 |
| Precision         | 0.7571 |
| Recall            | 0.8689 |
| Balanced Accuracy | 0.8935 |

## 4 RBF Kernel SVM

The Radial Basis Function (RBF) kernel allows the SVM to model more complex nonlinear decision boundaries. The kernel function is defined as

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

Two hyperparameters were tuned: the regularization parameter  $C \in [10^{-2}, 10^4]$  and the kernel width parameter  $\gamma \in [10^{-5}, 10^2]$ . The best parameters found were  $C = 26.827$  and  $\gamma = 0.01$ .

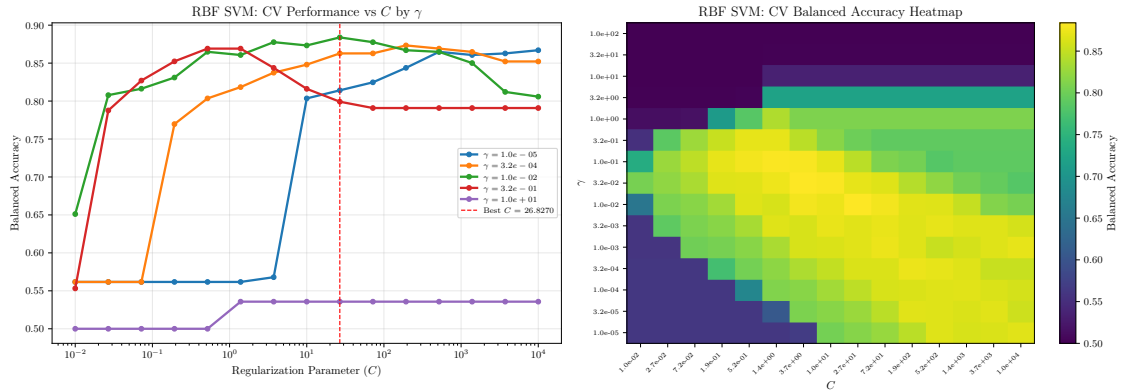


Figure 3: Cross-validation results for the RBF kernel across combinations of  $C$  and  $\gamma$ .

## Test Performance

Table 3: Test set performance of the RBF kernel SVM.

| Metric            | Value  |
|-------------------|--------|
| Accuracy          | 0.8603 |
| Sensitivity       | 0.9426 |
| Specificity       | 0.8361 |
| Precision         | 0.6284 |
| Recall            | 0.9426 |
| Balanced Accuracy | 0.8894 |

The RBF SVM achieved the highest sensitivity among all models, indicating excellent ability to detect DAT patients.

## Model Comparison

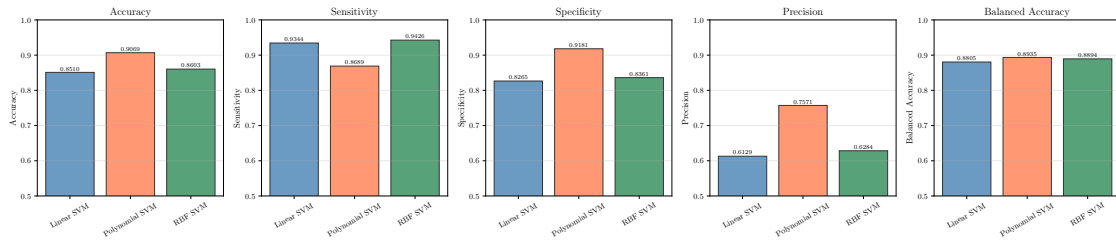


Figure 4: Comparison of performance metrics across linear, polynomial, and RBF SVM models.

Table 4: Summary of accuracy and balanced accuracy for all evaluated models.

| Model          | Accuracy | Balanced Accuracy |
|----------------|----------|-------------------|
| Linear SVM     | 0.8510   | 0.8805            |
| Polynomial SVM | 0.9009   | 0.8935            |
| RBF SVM        | 0.8603   | 0.8894            |

Among the tested approaches, the polynomial kernel SVM achieved the best overall performance.

## 5 Final Classifier

Based on the results from Sections 2–4, the polynomial SVM with  $C = 1.3895$  and  $d = 3$  was selected as the final classifier. This model achieved the highest balanced accuracy and overall classification accuracy among the evaluated models. To maximize predictive performance, the final classifier was retrained on all available labeled data prior to generating predictions for the unseen evaluation dataset.

## 6 Conclusion

This report evaluated three Support Vector Machine classifiers for distinguishing between stable normal controls and individuals with Dementia of Alzheimer’s Type using cortical glucose metabolism features. The experiments demonstrate that nonlinear kernels outperform the linear classifier. Among the tested models,

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the polynomial kernel SVM achieved the best balance between sensitivity and specificity and the highest overall classification accuracy. These results suggest that moderate nonlinear relationships exist between cortical metabolic activity and DAT diagnosis.